Sankaraleengam (Sankar) Alagapan

Georgia Institute of Technology
School of Electrical and Computer Engineering
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RESEARCH INTERESTS

Human Neuroscience Brain Stimulation, NeuroAI

EDUCATION

Ph.D. Biomedical Engineering 2014
University of Florida, Gainesville, FL

M.S. Biomedical Engineering 2008
University of Florida, Gainesville, FL

B.E. Electrical and Electronics Engineering 2007
Anna University, Chennai, India

RESEARCH EXPERIENCE

Research Scientist II 2022 –
Georgia Institute of Technology

Research Engineer II 2018 – 2022
Georgia Institute of Technology

Postdoctoral Research Associate 2014 – 2018
University of North Carolina at Chapel Hill

Graduate Research Assistant 2008 – 2014
University of Florida

HONORS & AWARDS

Georgia Institute of Technology
- Georgia CTSA KL2 Scholar 2023
- Best Faculty Paper Award – Sigma Xi 2024
- Roger. P Webb Outstanding Non-Tenure Faculty 2024

University of North Carolina at Chapel Hill
- Helen Lyng White Postdoctoral Fellowship 2016
- Office of Postdoctoral Affairs Travel Award 2016

University of Florida
- Office of Research Travel Award 2010, 2013
- Graduate Student Council Travel Award 2011
- College of Engineering Achievement Award 2007

RESEARCH SUPPORT

A naturalistic multimodal platform for capturing brain-body interactions in people during physical effort-based decision making 2025 - 2028

NIH NIMH R61MH138966
Role: Co-Investigator

Multimodal Biometrics to Optimize and Scale Decision Support for Depression DBS — 2025 - 2030
NIH NINDS UH3
Role: Co-Investigator

2023 – 2025
Georgia Clinical and Translational Science Alliance
NIH NCATS KLTR002381
Role: KL2 Scholar

Electrophysiological Biomarkers to Optimize DBS for Depression — 2019 - 2024
NIH NINDS UH3 NS103550
Role: Postdoc/ Team Lead

PUBLICATIONS (Google Scholar)

Journal Articles

Anand V., Helbling A., Davenport M., Berman G., **Alagapan S.**, Rozell C. (2026). LORE: Jointly Learning the Intrinsic Dimensionality and Relative Similarity Structure From Ordinal Data. *arXiv preprint arXiv:2602.04192.*

Pinotsis D. A., **Alagapan S.**, Sarikhani, P., Nauvel, T., Rozell, C. J., Mayberg, H. S. (2026). Ephaptic coupling and power fluctuations in depression. *Cerebral Cortex*, 36(3)

Desai A., Van Bogaert T., Choi K. S., Song H. N., Cha J., **Alagapan S.**, ... Waters A. C. (2025). Longitudinal Changes in Cortical Response Dynamics with Deep Brain Stimulation to the Subcallosal Cingulate for Treatment-Resistant Depression. bioRxiv, 2025-11.

Chen Y., Mudrik N., Johnsen K.A., **Alagapan S.**, Charles A.S. and Rozell C.J., (2025). Probabilistic decomposed linear dynamical systems for robust discovery of latent neural dynamics. *Advances in Neural Information Processing Systems*, 37, p 104443-104470.

Xu E., Pitts S., Dahill-Fuchel J., Scherrer, S., Nauvel T., …., **Alagapan S.**,…, Waters A.C., (2024). Neural Interoceptive Processing is Modulated by Deep Brain Stimulation for Treatment Resistant Depression. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, p S2451-9022.

Van Rheede J.*, **Alagapan S.***, Denison T., Riva Posse P., Rozell C., Mayberg H., Waters A.C., Sharrott A., Cortical signatures of sleep are altered following effective deep brain stimulation for depression. *Translational Psychiatry 14*:1

Alagapan S., Heisig S., Choi K., Riva Posse P., Crowell A., …., Mayberg H. M., and Rozell, C. (2023) Cingulate dynamics track depression recovery with deep brain stimulation *Nature* 622, pages 130–138

Smith E.E., Choi K.S., Veerakumar A., Obatusin M., Howell B., Smith A.H., Crowell A., Riva Posse P., **Alagapan S.**, Rozell C.J., Mayberg H.S., and Waters A.C. (2022) Time-frequency signatures evoked by single-pulse deep brain stimulation to the subcallosal cingulate. *Frontiers in Human Neuroscience, 16:939258*

Wong, J. K., Deuschl, G., Wolke, R., Bergman, H., Muthuraman, M., Groppa, S., ... **Alagapan, S.**, … and Okun, M. S. (2022). Proceedings of the Ninth Annual Deep Brain Stimulation Think Tank: Advances in Cutting Edge Technologies, Artificial Intelligence, Neuromodulation, Neuroethics, Pain, Interventional Psychiatry, Epilepsy, and Traumatic Brain Injury. *Frontiers in Human Neuroscience*, 16:813387

Alagapan S., Riddle, J., Huang, W.A., Hadar, E., Shin H.W., Fröhlich., (2019) Network-targeted, multi-site direct cortical stimulation enhances working memory by modulating phase lag of low frequency oscillations. *Cell Reports* 29 (9), 2590-2598

McPherson, T. †, Berger, D.S., **Alagapan, S.**, and Fröhlich, F. (2019) Active and passive music therapy interventions differentially modulate sympathetic autonomic nervous system activity. *Journal of Music Therapy* 56 (3), 240-264

Alexander, M.L.*, **Alagapan, S.***, Lugo, C.E., Mellin, J.M., Lustenberger, C., Rubinow, D.R., and Fröhlich, F. (2019) Double-blind, randomized pilot clinical trial targeting alpha oscillations with transcranial alternating current stimulation (tACS) for the treatment of major depressive disorder (MDD). *Translational Psychiatry* 9:106

Alagapan, S., Fröhlich, F., and Wu, H-T., Diffusion geometry approach to efficiently remove electrical stimulation artifacts in intracranial electroencephalography. (2018) *Journal of Neural Engineering* 16, 036010

Sheffield, A. †, Ahn, S., **Alagapan, S.**, and Fröhlich, F. (2018) Investigating Effects of Neural Oscillations by Lateral Transcranial Static Magnetic Field Stimulation on the Prefrontal Cortex: A crossover, double-blind, sham-controlled pilot study. *European Journal of Neuroscience* 00:1–13.

Ahn S., Mellin, J.M., **Alagapan, S.**, Lugo, C.E., Alexander, M.L., Gilmore, J.H., Jarskog, L.F., and Fröhlich, F. (2018) Targeting impaired neural oscillations in patients with schizophrenia by transcranial alternating current stimulation: A randomized, double-blind, placebo-controlled pilot study. *NeuroImage* 186, 126-136

Alagapan, S., Lustenberger, C., Hadar, E., Shin, H.W., and Fröhlich, F. (2018) Low-frequency direct cortical stimulation of left superior frontal gyrus enhances working memory performance. *NeuroImage* 184, 697-706

McPherson, T. †, Berger, D.S., **Alagapan, S.**, and Fröhlich, F. (2018) Intrinsic Rhythmicity Predicts Synchronization-Continuation Entrainment Performance. *Scientific Reports* 8, 11782

Mellin, J.M., **Alagapan, S.**, Lustenberger, C., Lugo, C.E., Alexander, M.L., Gilmore, J.H., Jarskog, L.F., and Fröhlich, F. (2018). Randomized Trial of Transcranial Alternating Current Stimulation for Treatment of Auditory Hallucinations in Schizophrenia. *European Psychiatry* 51, 25-33.

Lustenberger, C., Patel, Y.A., **Alagapan, S.**, Page, J.M., Price, B., Boyle, M.R., and Fröhlich, F. (2017). Topographical characterization of brain responses to auditory rhythmic stimuli during wakefulness and NREM sleep. *NeuroImage* 169, 57-68

Sellers, K.K., Yu, C.X., Zhou, Z.C., Stitt, I., Li, Y.H., Radtke-Schuller, S., **Alagapan, S.**, and Fröhlich, F. (2016). Oscillatory Dynamics in the Frontoparietal Attention Network during Sustained Attention in the Ferret. *Cell Reports* 16, 2864-2874.

Lustenberger, C., Boyle, M.R., **Alagapan, S.**, Mellin, J.M., Vaughn, B.V., and Fröhlich, F. (2016). Feedback-Controlled Transcranial Alternating Current Stimulation Reveals a Functional Role of Sleep Spindles in Motor Memory Consolidation. *Current Biology* 26, 2127-2136.

DeMarse, T.B., Pan, L., **Alagapan, S.**, Brewer, G., and Wheeler, B.C. (2016). Feed-Forward Propagation of Temporal and Rate Information between Cortical Populations during Coherent Activation in Engineered In Vitro Networks. *Frontiers in Neural Circuits* 10.

Alagapan, S., Schmidt, S.L., Lefebvre, J., Hadar, E., Shin, H.W., and Fröhlich, F. (2016). Modulation of Cortical Oscillations by Low-Frequency Direct Cortical Stimulation Is State-Dependent. *PLoS Biology* 14, e1002424.

Alagapan, S., Franca, E., Pan, L., Leondopulos, S., Wheeler, B.C., and DeMarse, T.B. (2016). Structure, Function, and Propagation of Information across Living Two, Four, and Eight Node Degree Topologies. *Frontiers in Bioengineering and Biotechnology* 4, 15.

Franca, E., Jao, P.F., Fang, S.P., **Alagapan, S.**, Pan, L., Yoon, J.H., Yoon, Y.K., and Wheeler, B.C. (2016). Scale of Carbon Nanomaterials Affects Neural Outgrowth and Adhesion. *IEEE Transactions in Nanobioscience* 15, 11-18.

Pan, L.*, **Alagapan, S.***, Franca, E., Leondopulos, S.S., DeMarse, T.B.*, Brewer, G.J., and Wheeler, B.C. (2015). An in vitro method to manipulate the direction and functional strength between neural populations. *Frontiers in Neural Circuits* 9, 32.

Pan, L., **Alagapan, S.**, Franca, E., DeMarse, T., Brewer, G.J., and Wheeler, B.C. (2014). Large extracellular spikes recordable from axons in microtunnels. *IEEE Transactions in Neural Systems and Rehabilitation Engineering* 22, 453-459.

Brewer, G.J., Boehler, M.D., Leondopulos, S., Pan, L., **Alagapan, S.**, DeMarse, T.B., and Wheeler, B.C. (2013). Toward a self-wired active reconstruction of the hippocampal trisynaptic loop: DG-CA3. *Frontiers in Neural Circuits* 7, 165.

Pan, L., **Alagapan, S.**, Franca, E., Brewer, G.J., and Wheeler, B.C. (2011). Propagation of action potential activity in a predefined microtunnel neural network. *Journal of Neural Engineering* 8, 046031.

Book Chapters

Fröhlich, F., **Alagapan, S.***, Boyle, M.R.*, Hamilton, F.*, Li, G.*, Lustenberger, C.*, and Schmidt, S.L.* (2016). Target Engagement with Transcranial Current Stimulation. In *Transcranial Direct Current Stimulation in Neuropsychiatric Disorders: Clinical Principles and Management*, A. Brunoni, M. Nitsche and C. Loo, eds., pp. 197-222.

* Contributed equally † Mentee

TALKS

CNS 2025 Workshop, Florence, Italy	2025
Minnesota Neuromodulation Symposium, Minneapolis, MN	2025
6th Bioelectronic Medicine Summit, New York, NY	2025

Neurology Grand Rounds, Emory University, Atlanta, GA	2024
O'Donnell Brain Institute, UT Southwestern, Dallas, TX	2024
Psychology Dept Seminar, Emory University, Atlanta, GA	2023
Society for Neuroscience Annual Meeting (Nanosymposium), Washington DC	2023
Society for Neuroscience Annual Meeting (Nanosymposium), San Diego, CA	2022
Society of Biological Psychiatry Annual Meeting (Symposium), New Orleans, LA	2022
Transcranial Electric Stimulation Workshop (Carolina Neurostimulation Conference), Chapel Hill, NC	2018
Society for Industrial and Applied Mathematics Southeast Atlantic Section Conference, Chapel Hill, NC	2018
Neuroscience Center Seminar Series, UNC Chapel Hill, NC	2017

TEACHING & MENTORING

Teaching

University of North Carolina at Chapel Hill

Guest Lecturer, Network Neuroscience	2015, 2017

University of Florida

Graduate Teaching Assistant, Neural Engineering	2011
Graduate Teaching Assistant, Neurodynamics	2010

Mentoring

Georgia Institute of Technology

Postdoctoral Scholars:

Parisa Sarikhani,	2024 –
Ermyntrude Adjei	2026 –

Graduate Students:

Elif Ceren Fitoz, ECE, PhD	2022 –
Nayanika Biswas, ECE, MS	2022
Tyler Albarran, ECE, PhD	2023 –
Vivek Anand, ECE, PhD	2024 –
Shreesh Karjagi, ECE, PhD	2025 –
Dong Yeun Kim, ME, MS	2025 –

University of North Carolina at Chapel Hill

Jhana Parikh, Undergraduate research assistant (Honors Thesis Committee)	2018
Trevor Mcpherson, Undergraduate research assistant	2017 – 2018
Florian Schertenleib, Visiting graduate researcher	2016 – 2017

	Matt Mattoni, Undergraduate research assistant	2016 – 2017

SERVICE	Review Panelist, National Science Foundation	2025
	Reviewer, MEXA Accelerator,	2025
	Nanosymposium Chair, Society for Neuroscience Annual Meeting	2022
	Conference Program Committee - Carolina Neurostimulation Conference	2018
	Ad-hoc Reviewer Nature, Science, Nature Communications, Biological Psychiatry, Biological Psychiatry: Cognitive Neuroscience and Neuroimaging, Journal of Computational Neuroscience, Psychological Medicine, Scientific Reports, Psychiatry Research, Neuroscience Letters, Journal of Neuroscience Methods, NeuroImage, Frontiers in Neuroscience, IEEE Journal of Biomedical and Health Informatics	2016 –